

SYNOPSIS



At approximately 0600 on 3 October 2007, *Lady Hamilton of Helford* left the Helford River and headed across Falmouth Bay at 6 knots towards the 'Old Wall' fishing grounds. Her skipper was on watch in the wheelhouse while the vessel's two deckhands slept. At approximately 0615, *Blithe Spirit* left Falmouth and headed south towards the Manacles at a speed of between 18 and 26 knots. Shortly after leaving the harbour, the skipper of *Blithe Spirit*, who was operating single handed, altered course to port to avoid a cluster of unidentified lights ahead of him. Once steady on his new course, with the way ahead apparently clear, he sat down to rest.

At approximately 0622, *Blithe Spirit* and *Lady Hamilton of Helford* collided. The crew of *Lady Hamilton of Helford* were thrown to the deck and covered in debris. *Lady Hamilton of Helford* was holed above and below the waterline and started to rapidly take on water. Her skipper used a mobile phone to inform the coastguard while his crew launched a liferaft. The liferaft did not fully inflate and was unusable. Fortunately, a local pilot boat quickly arrived on the scene and recovered the skipper and his crew. The damaged vessel was taken in tow but sank at 0646. When the two vessels collided, the skipper of *Blithe Spirit* hit his head on a chart plotter on impact before landing on the deck of his boat. It would appear that he then lost consciousness for a short period. After coming to, *Blithe Spirit's* skipper took the boat back to Falmouth, arriving there at about 0715.

Neither skipper saw the other vessel immediately before or after the collision and both concluded they had struck semi-submerged objects. The investigation identified a number of factors which led to the collision. In particular, neither skipper maintained an effective lookout.

To improve fishing vessel safety, the MCA intends to require skippers of vessels under 16.5m to hold a certificate of competency and is considering the provision of a team of safety advisors to raise awareness of the risks associated with a fishing vessel's routine operation. It is also considering the promulgation of a minimum recognised standard for liferafts carried on board fishing vessels of less than 15m.

The investigation also identified that the liferaft deployed from *Lady Hamilton of Helford* had not been serviced in accordance with its manufacturer's instructions and did not fully inflate because there was insufficient gas in its cylinder. As a result, the MAIB strongly advises that all owners of leisure craft and fishing vessels who own or hire liferafts seek confirmation from service agents that all maintenance, checks and tests conducted have been in accordance with the manufacturer's instructions.

Recommendations have been made to the MCA and Premium Liferaft Services, which seek to improve the reliability of liferafts carried on smaller vessels. A further recommendation to the MCA aims to improve the safe operation of fishing vessels capable of fast speeds.

1.2 NARRATIVE

1.2.1 *Lady Hamilton of Helford*

At approximately 0535 on 3 October 2007, the skipper and two deckhands of *Lady Hamilton of Helford* met on the beach at Helford. After making their way to the gill-netter, they loaded supplies and equipment onto the boat and carried out their usual pre-sailing preparations. The skipper turned on the navigation and deck lights, started the engine and pumped the vessel's bilge; the deckhands washed the bridge windows and prepared the deck for fishing.

Lady Hamilton of Helford was cast off from her mooring on the Helford River at about 0600 and headed east into Falmouth Bay for passage to the 'Old Wall' fishing grounds south of St Anthony Head, at a speed of 6 knots (**Figure 1**). It was dark, and during the passage one deckhand went to sleep on a bench in the cuddy forward of the wheelhouse; the other slept on the engine casing in the wheelhouse.

The skipper used a chart plotter to monitor the vessel's position, and kept a visual lookout from the steering position on the starboard side of the wheelhouse. The starboard forward wheelhouse window was opened to make it easier to spot crab pot marker buoys laid by other fishermen. The skipper occasionally leant through the open window and looked to port for other vessels heading south from Falmouth harbour, but none were seen.

As he approached the area where he intended to shoot his first set of nets, the skipper's attention was divided between avoiding fishing buoys and watching his chart plotter. At approximately 0622, *Lady Hamilton of Helford's* port side made heavy contact with an object.

The impact threw the skipper to the deck and the vessel went into a starboard turn. The wheelhouse structure was displaced to starboard, and what remained of the wheelhouse was covered in debris. The deckhand in the cuddy made his way through the debris into the wheelhouse where the skipper found the other deckhand under the engine casing which had collapsed. An assessment of the damage established that the vessel was holed and was taking in water.

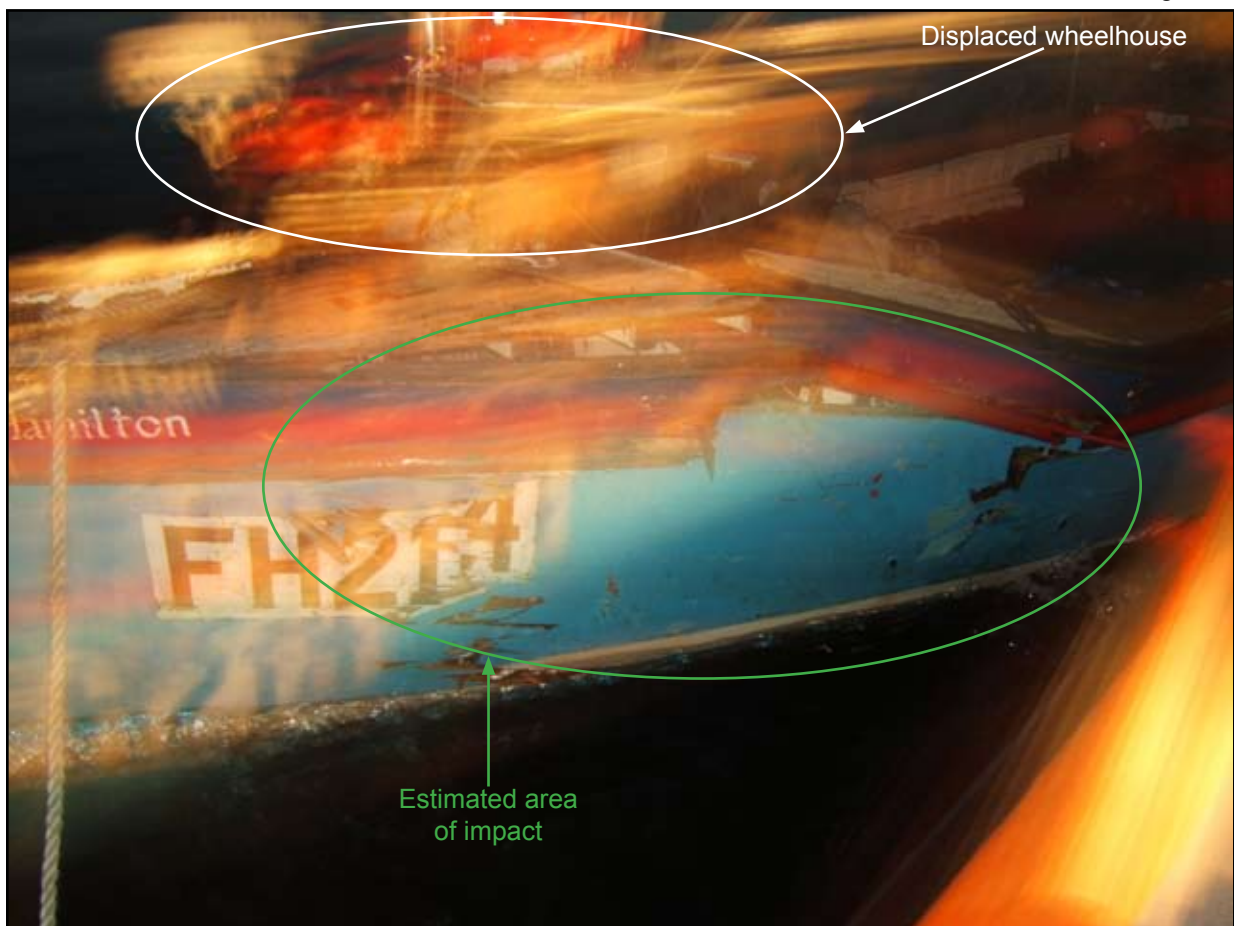
The water level within the boat rose quickly and it was apparent that the vessel was sinking. About 1 minute after the impact, the skipper told the deckhands to look for other craft in the immediate vicinity while he alerted the coastguard using a mobile telephone. The skipper was unable to access the VHF radio, flares or bilge pump. No other vessels were seen nearby and the skipper assessed that *Lady Hamilton of Helford* had hit a floating or semi-submerged object. As the vessel continued under helm to starboard, power was lost to her navigation and deck lights.

Prompted by the coastguard, the crew found and put on their lifejackets. The skipper then located the engine control panel among the debris and stopped the engine. This enabled the deckhands to launch the liferaft. Once it was lowered into the water, its activation cord was pulled but, although its canister opened, the raft did not fully inflate. The crew considered the liferaft might have inflated upside down and turned it over, but this was not the case.

On receipt of the call from the skipper of *Lady Hamilton of Helford*, the coastguard broadcast a “Mayday” and requested the assistance of vessels in the area. At 0628, it also tasked the RNLi inshore and all weather lifeboats. The pilot cutter, *Arrow*, responded to the “Mayday” and came alongside the stricken fishing vessel at 0632. The coxswain of *Arrow* had initially headed towards two radar targets. As he closed, he saw the mizzen sail of *Lady Hamilton of Helford*, which he associated with the southerly of the targets. The coxswain assumed that the second radar target, which was about 1 mile further to the north, was a fisherman’s marker buoy.

The two deckhands transferred to *Arrow* at 0634 while the skipper attached a line to the bow cleat of his vessel. He then also transferred to *Arrow*, and *Lady Hamilton of Helford* was taken in tow to the north east, towards Maenporth. However, after being towed for less than 200m, she foundered at 0646 in an upright attitude in position 50° 06.41’N 005° 02.59’W. A photograph of the vessel taken on a digital camera moments before she sank is at **Figure 2**. The crew of *Lady Hamilton of Helford* were transferred to the inshore lifeboat at 0655.

Figure 2



Photograph taken of *Lady Hamilton of Helford* before she foundered

The coastguard tug *Anglian Princess* was tasked to assist in the search for other boats and any floating objects that might have been struck by *Lady Hamilton of Helford*. At 0654, an intermittent red light was seen, which associated with a radar target 1.2 miles to the north of the foundered vessel. The red light was again seen 1.6 miles north of the scene at 0659, but was not observed thereafter. A piece of timber approximately 3m in length was recovered from the vicinity of the impact at 0749.

1.2.2 *Blithe Spirit*

The dory, *Blithe Spirit*, left Grove slipway in Falmouth harbour between 0600 and 0615. Her skipper, working alone, transited the harbour at a speed of between 8 and 10 knots, but increased speed to between 18 and 26 knots as he entered open water. He intended to head south to fish for sea bass around a group of rocks known as the Manacles (**Figure 1**).

Soon after leaving the harbour, the skipper saw a number of lights about 3 miles ahead when standing at the steering position. He was able to identify two large ships but was uncertain about the source of a third group of lights. The lights did not seem to be moving, and the skipper assessed that they were just to the west of his intended track. With the intention of increasing the passing distance of the unidentified lights to about half a mile, the skipper altered course several degrees to port. With the way ahead apparently clear he then sat down; he was not wearing a kill-cord¹ or a lifejacket.

The next recollection the skipper had was of finding himself lying in the bottom of the dory, which was now moving at about 6 knots through the water, with its engine running at much reduced throttle. He did not remember an impact, but quickly realised that his boat had hit something. The skipper stood up, put the engine in neutral, and looked to see if he could see any other vessel or object nearby. He could not. The skipper was feeling very groggy and had to squint to focus. He then carried out a slow turn to port and headed back to Falmouth harbour at a speed of between 10 and 15 knots. He assumed that *Blithe Spirit* had hit a log or other floating debris released during the recent demolition of a pier.

Blithe Spirit returned to Grove slipway at about 0715, and her skipper was helped ashore by two yachtsmen who saw that he had sustained injuries to his face. The yachtsmen made the skipper aware of the seriousness of his injuries and persuaded him that he needed medical treatment. They also informed him of the accident to *Lady Hamilton of Helford*. The yachtsmen moved *Blithe Spirit* onto the slipway and then took her skipper to Falmouth hospital. Aware of the loss of *Lady Hamilton of Helford* and the damage to *Blithe Spirit*, the skipper realised that there was a possibility that the two vessels had collided, and reported his accident to the coastguard. The skipper was later transferred to Truro hospital where he was treated for facial injuries and a bruised knee.

1.3 ENVIRONMENTAL CONDITIONS

At the time of the collision, it was dark, the wind was southerly force one, the sea was calm to slight and the visibility was good. Although there was some mist reported in the area, none was seen in the vicinity of the accident. Predicted high water at Falmouth was at 1027. Civil twilight² was at 0651 and sunrise was at 0722.

¹ Kill-cords connect the helmsman to an engine shut down switch and are designed to stop the engine if he is thrown to the deck or overboard.

² In the morning, civil twilight is the period between when the sun is 6 degrees below the horizon and sunrise. Depending on cloud cover, there is usually enough light from the sun to conduct outdoor activities during this period, without the aid of artificial light sources.

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 PREVIOUS INCIDENTS

2.2.1 The vessels

In 2001, *Lady Hamilton of Helford* was involved in a near miss incident with the container ship *Mathilda*, which was subject to an MAIB investigation. The container ship failed to take avoiding action while on a collision course with *Lady Hamilton of Helford*, but the crew of the fishing vessel managed to cut their fishing gear and manoeuvred clear. The MAIB investigation report concluded that the master of *Mathilda* had not maintained a proper lookout.

Blithe Spirit was also involved in a near miss incident in 2003 when the skipper, again on passage to the Manacles, passed across the bows of a large cargo ship which he had mistakenly assessed to have been stationary. The skipper's manoeuvre was seen by the crew of a pilot boat and was reported to the harbourmaster in Falmouth.

2.2.2 Liferafts

The circumstances of the failure of a liferaft contained in Ireland's Marine Casualty Investigation Board (MCIB) investigation report into the sinking of the yacht *Megawatt*, are similar in a number of ways to this incident:

- The liferaft failed to fully inflate
- The liferaft was on hire
- The liferaft was in date for inspection/service
- The liferaft was not SOLAS compliant
- The gas had leaked past the loosely connected operating head
- The crew were rescued by another yacht before *Megawatt* sank.

The MCIB report made two recommendations relating to the failure of the liferaft:

1. *The Maritime Safety Directorate should issue a Marine Notice recommending non-SOLAS/MED⁸ liferafts to be serviced only by authorised agents appointed by the manufacturer.*
2. *The Marine Survey Office should investigate the servicing of non-MED liferafts and introduce legislation in relation to servicing if considered necessary.*

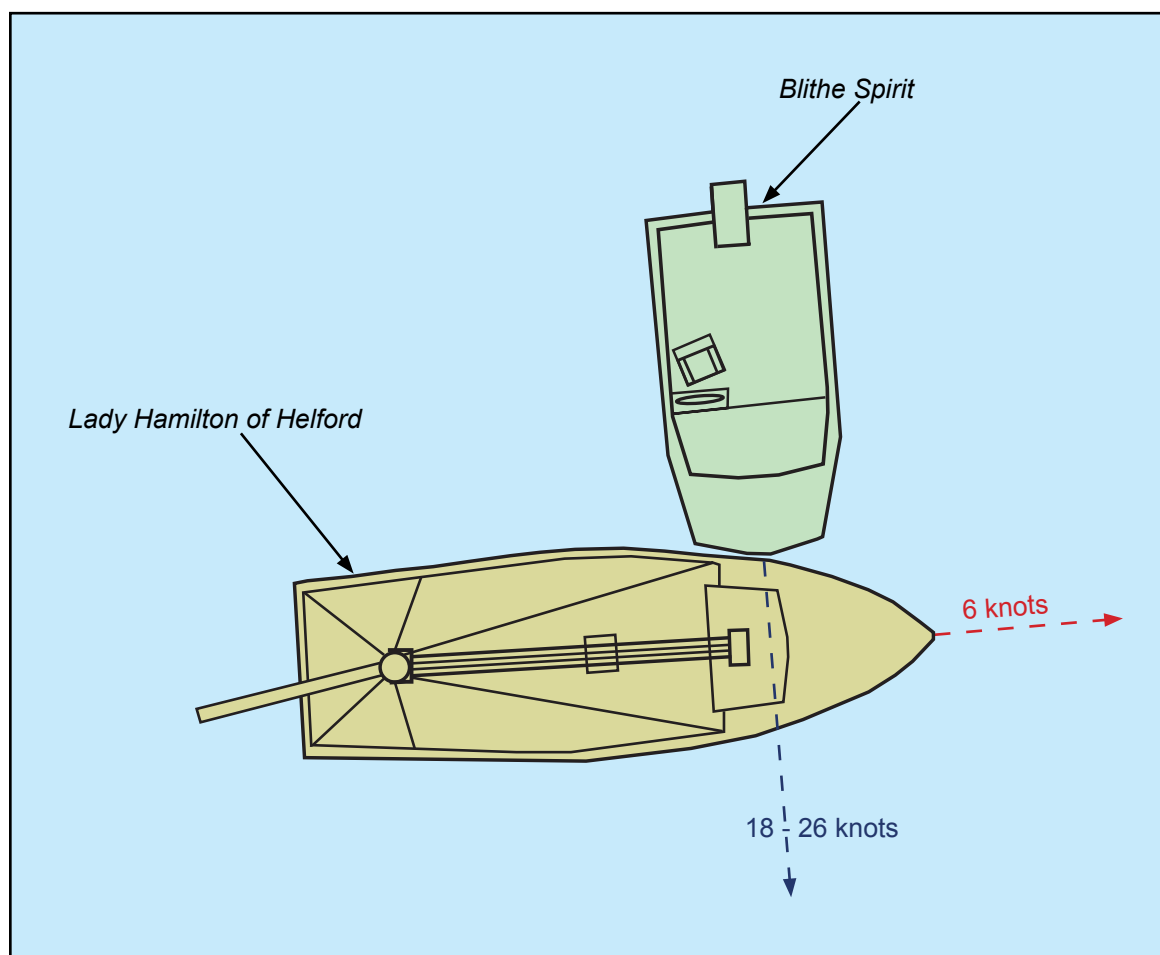
⁸ The MED (Marine Equipment Directive) is an EU Directive which was published in December 1996 and was adopted into UK law in 1999 (Statutory Instrument 1999 N° 1957 The Merchant Shipping (Marine Equipment) Regulations). The MED covers any vessel flying the flag of an EEA member state. Compliance of marine equipment with the MED is shown by the ship's wheel mark.

2.3 THE COLLISION

As the skippers of both vessels did not see another vessel in close proximity, either before or after the collision, their assessments that they had hit semi-submerged objects was understandable. However, notwithstanding the small piece of timber later found nearby, it is evident from the damage sustained (**Figures 2 and 5 to 9**) and post-accident surveys that the vessels collided with each other when both were making way. First, the paint found on the bow and chines of *Blithe Spirit* matched the paint on the hull of *Lady Hamilton of Helford*. Second, the windows found on the deck of *Lady Hamilton of Helford* matched the windows missing from *Blithe Spirit*. Third, the stem of *Lady Hamilton of Helford* was undamaged (**Figure 5**). Finally, the extent of the damage to *Lady Hamilton of Helford* would probably have required a speed of impact much faster than 6 knots.

Lady Hamilton of Helford was struck on her port side forward and the damage to *Blithe Spirit* indicates she impacted with her starboard bow. The relative positions of the damage and the force of impact is consistent with the directions and the speeds of the vessels at the time (**Figure 13**). The geographical location of the collision is also consistent with intersection of the vessels' intended passages. The paint marks on the starboard and centre chines of *Blithe Spirit* (**Figure 9**) indicate that the dory rode up the side of *Lady Hamilton of Helford* before deflecting to port.

Figure 13



Estimated relative position of vessels on impact

2.4 LOOKOUT

2.4.1 Regulation

It is a fundamental requirement of the COLREGS that vessels maintain a proper lookout. If they do not, many of the regulations intended to avoid collisions in varying circumstances cannot be applied. The importance of keeping a proper lookout is also emphasised to fishermen in MGN 313(F). As neither skipper saw the other vessel, despite the good visibility and calm conditions, it is apparent that an effective lookout was not being kept immediately before the collision.

2.4.2 *Blithe Spirit*

It is highly probable that the lights shown by *Lady Hamilton of Helford* were among those initially seen by the skipper of *Blithe Spirit*. The brightness of the gill-netter's deck lights would have made it difficult to distinguish the vessel's navigation lights. Therefore, it was not surprising that *Blithe Spirit*'s skipper was unable to determine the direction in which the gill-netter was travelling. However, having taken action to avoid the lights to the south by half a mile, he then sat down in a position from where his view ahead was extremely restricted. Consequently, he did not monitor the effectiveness of his course alteration or make any further attempts to assess the movement of the lights ahead. It is not certain how long the skipper was seated, but when travelling at between 18 and 26 knots, it would only have taken between 7 and 10 minutes to cover the 3 mile distance the skipper had estimated the lights to be from his vessel. Considering the potential difficulty in judging distances during darkness, it is possible *Lady Hamilton of Helford* was much closer than the skipper assessed, and the time interval from the course alteration to the collision was even shorter. Regardless of the time the skipper of *Blithe Spirit* was seated, it is evident that the lookout routine he adopted was not commensurate with the speed of his vessel or the absence of other navigational aids such as radar to determine the range and movements of other vessels.

2.4.3 *Lady Hamilton of Helford*

The ability of *Lady Hamilton of Helford*'s skipper to detect *Blithe Spirit* was impeded by the combination of several factors. First, the glare from the bright deck lights would have adversely affected his night vision. Second, the vessel's radar was not switched on. Third, the lights displayed by *Blithe Spirit* would have merged among the shore lights from Falmouth and St Mawes to the north. Fourth, as the skipper found it necessary to lean through a forward facing window to get an unobstructed view to port, the view to port from the steering position was not as clear. Fifth, his inability to distinguish between green and white lights would have made the approach of a vessel from the north less obvious. Finally, the skipper was focussed on avoiding fishing marks ahead and monitoring the chart plotter to ensure his vessel was in the correct position to shoot her nets.

As the bulb of *Blithe Spirit*'s all round white light was subsequently found to have failed, and the bulb in the starboard sidelight, which was an auto bulb and not the correct type for its housing, had been dislodged, it is also possible that one or both of these lights were not illuminated. However, it is equally possible that the sidelight bulb was dislodged and the all round light bulb failed when the vessels collided.

2.5 COMPLACENCY

Both skippers had a good knowledge of the local area and had completed the mandatory training courses required to operate their vessels. However, they failed to keep an effective lookout despite both having previous experience of near misses (paragraph 2.2.1). The skippers' actions inevitably raise concerns regarding competency, particularly as there is currently no requirement for skippers of fishing vessels of less than 16.5m length overall to hold a certificate of competency. However, both were working repetitive routines in a familiar and comfortable environment, and it is more likely that their behaviour was the result of an acceptance of risk, rather than a lack of knowledge of the COLREGS. The skippers had worked to their normal working patterns during the week leading to the incident, and were considered to have been well rested the night before the collision. Therefore, fatigue is unlikely to have been a contributory factor on this occasion.

It was dark, difficult to detect the lights of vessels among the shore lights to the north, and the workload of the skipper of *Lady Hamilton of Helford* increased as he approached the fishing grounds. However, this was a situation he experienced on a daily basis and it is evident from his decision not to switch on the radar unless the visibility was poor and to allow his crew to sleep, that he was not concerned by the increased risk to his vessel. Moreover, by keeping his deck lights on when underway, it is possible he assumed that she would be seen by other vessels approaching from the north which would then be obliged by the COLREGS to keep out of his way. However, such an assumption is reliant on other vessels keeping a proper lookout and taking the appropriate action. This accident clearly demonstrates that this is not always the case. Had the skipper used his radar, and employed one of his crew to act as lookout, there is a high probability that the fast moving *Blithe Spirit* would have been detected in sufficient time for avoiding action to have been taken.

The actions of *Blithe Spirit's* skipper also indicate a complacent approach to the operation of his craft. He habitually transited the bay at speed during darkness and was fully aware that he was unable to see ahead when seated. However, he made no attempt to monitor the effectiveness of his alteration to port. The skipper's lack of action in this respect not only indicates an acceptance or disregard of the risks involved, but it also demonstrates a lack of awareness of the pace at which situations change when travelling at speed. The poor condition of the pyrotechnics on board *Blithe Spirit*, and her skipper's decision not to wear a lifejacket or kill-cord are further examples of the skipper's casual attitude to safety. On this occasion, he was extremely fortunate. Had he been thrown into the water rather than into the boat on impact, it is almost certain that, unconscious and without a lifejacket, he would have drowned.

2.6 EVENTS AFTER THE COLLISION

It is not possible to determine at what speed, or in which direction, *Blithe Spirit* left the scene of the collision but, by the time the crew of *Lady Hamilton of Helford* had stood up and looked around, she could have been a considerable distance away. Furthermore, it was still dark and, with *Blithe Spirit's* all round white light and starboard sidelight extinguished, she would not have been readily visible on anything but a port aspect. It would also have been difficult to hear *Blithe Spirit's* relatively quiet four stroke outboard engine over the noise produced by the larger diesel engine of *Lady Hamilton of Helford*, which was also under constant helm to starboard.

SECTION 3 - CONCLUSIONS

3.1 SAFETY ISSUES WHICH HAVE RESULTED IN RECOMMENDATIONS

1. The actions of the skipper of *Blithe Spirit* indicate his lack of awareness of the pace at which situations change when travelling at speed. [2.5]
2. The liferaft service station did not follow the manufacturer's service requirements for the Lifeguard Forties liferaft, and the manufacturer's accreditation of the service station for this liferaft type had lapsed. [2.7.1]
3. A greater assurance of the liferaft's reliability would have been afforded had the liferaft been subject to similar service regimes to those required for the service station's SOLAS compliant and privately owned liferafts. [2.7.1]
4. There are a number of liferaft defects which might not be identified without full-system gas inflation. [2.7.1]

3.2 SAFETY ISSUES WHICH HAVE NOT RESULTED IN RECOMMENDATIONS BUT HAVE BEEN ADDRESSED

1. Neither the skipper of *Lady Hamilton of Helford* nor the skipper of *Blithe Spirit* maintained an effective lookout. [2.4]
2. In view of their repetitive routine in a familiar environment, it is likely that the actions of the skipper of *Lady Hamilton of Helford* and the skipper of *Blithe Spirit* was the result of an acceptance of risk, rather than a lack of knowledge of the COLREGS. [2.5]
3. Recent trials conducted by the RNLI indicated that the performance of liferafts which are not manufactured to a recognised standard was questionable. [2.7.2]
4. In the absence of a stipulated or recommended recognised standard for liferafts carried on board fishing vessels of less than 10m LOA, it is highly likely that many fishermen will opt for the least expensive and less reliable options if not advised or directed to do otherwise. [2.7.2]